



資料結構

Data Structure

Lab (加分題)

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Lab_bonus-Ex1. (heaters)

完成 leetcode-Heaters

Code

```
class Solution {
public:
    int findRadius(vector<int>& houses, vector<int>& heaters) {
        vector<int> hk;
        int max;
        int r;
        int a = houses.size(); //設定a為房子數量
        int b = heaters.size(); //設定b為加熱器數量
        for(int x = 0 ; x<a ;x++) { //巢狀迴圈 內層是房子到每一個加熱器的最大距離
            r= abs(houses[x]-heaters[0]);
            for(int y=0 ;y<b ;y++){
                if(r>abs(houses[x]-heaters[y])){
                    r = abs(houses[x]-heaters[y]);
                }
            }
            hk.push_back(r); //總共有a個元素在hk陣列=房子數量
        }
        max = hk[0];
        for(int z = 0 ; z<hk.size();z++) { //這裡是取最大的r
            if(hk[z]>max){
                max = hk[z];
            }
        }
        return max; //回傳最大的r
    }
}
```

Result

The screenshot displays the LeetCode interface for the 'Heaters' problem. The submission is marked as 'Accepted' with 30/30 test cases passed. Performance metrics show a runtime of 1548 ms (Beats 5.80%) and memory usage of 30.22 MB (Beats 5.12%). A bar chart illustrates the runtime distribution across various test cases. The code is written in C++ and includes detailed comments in Chinese explaining the logic: calculating the distance from each house to the nearest heater and then finding the maximum of these distances. The test result section shows three cases, all of which were passed successfully.

leetcode.com/problems/heaters/

Accepted 30 / 30 testcases passed
114310123 submitted at May 27, 2026 21:43

Runtime: 1548 ms | Beat: 5.80%
Memory: 30.22 MB | Beat: 5.12%

0.25% of solutions used 271 ms of runtime

Code | C++

```
1 class Solution {
2 public:
3     int findRadius(vector<int>& houses, vector<int>& heaters) {
4         vector<int> hk;
5         int max;
6         int a = houses.size(); // 設定 a 為房子數量
7         int b = heaters.size(); // 設定 b 為加熱器數量
8         for(int x = 0; x < a; x++) { // 遍歷房子到每一個加熱器的最大距離
9             r = abs(houses[x] - heaters[0]);
10            for(int y = 0; y < b; y++) {
11                if(r > abs(houses[x] - heaters[y])) {
12                    r = abs(houses[x] - heaters[y]);
13                }
14            }
15            hk.push_back(r); // 總共有 a 個元素在 hk 陣列-房子數量
16        }
17        max = hk[0];
18        for(int z = 0; z < hk.size(); z++) { // 這裡是取最大的 r
19            if(hk[z] > max) {
20                max = hk[z];
21            }
22        }
23        return max; // 回傳最大的 r
24    }
25 }
26 }
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

houses = [1,2,3,4]

heaters =

leetcode.com/problems/heaters/

Accepted 30 / 30 testcases passed
114310123 submitted at May 27, 2026 21:43

Runtime: 1548 ms | Beat: 5.80%
Memory: 30.22 MB | Beat: 5.12%

0.19% of solutions used 336 ms of runtime

Code | C++

```
1 class Solution {
2 public:
3     int findRadius(vector<int>& houses, vector<int>& heaters) {
4         vector<int> hk;
5         int max;
6         int r;
7         int a = houses.size();
8         int b = heaters.size();
9         for(int x = 0; x < a; x++) { // 設定 a 為房子數量
10            for(int y = 0; y < b; y++) { // 設定 b 為加熱器數量
11                r = abs(houses[x] - heaters[y]);
12                if(r > abs(houses[x] - heaters[0])) {
13                    r = abs(houses[x] - heaters[y]);
14                }
15            }
16            hk.push_back(r); // 總共有 a 個元素在 hk 陣列-房子數量
17        }
18        max = hk[0];
19        for(int z = 0; z < hk.size(); z++) { // 這裡是取最大的 r
20            if(hk[z] > max) {
21                max = hk[z];
22            }
23        }
24        return max; // 回傳最大的 r
25    }
26 }
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

houses = [1,5]

heaters =

Discussion

這次的加分題較簡單,我沒有用半點ai就完成 很有成就感,
但是我使用了巢狀迴圈暴力破解法讓我的時間複雜度變成 $O(n^2)$
這很不好,感覺可以透過房子座標往左和右邊推直到遇到heater之類的思維來優化
但是我有點累所以就用暴力破解

$x = \text{houses point}$

$y = \text{heaters - point}$



